Launch Analog and Input Tool: User Journey

Motivating question: How many TRx will a new brand sell in its first three years, how uncertain is that, and what drives the uncertainty?

Purpose: Help a pharma forecasting team choose and defend the two inputs a launch forecast depends on: how big the launch can get (the ceiling) and how fast it gets there (the trajectory). The tool recommends comparable launches, sizes the market, and shows an illustrative curve with its limitations stated.

Users

- **Primary user:** forecaster (enters the profile, reviews and approves analogs, sets the funnel assumptions, defends the inputs).
- **Audience for results:** brand leadership and finance, who see the analog set, the ceiling and trajectory assumptions, and the illustrative curve through the memo and final presentation. They use these to judge whether the inputs are defensible and what forecast to commission.

Fixed assumptions (not user inputs): US only; new molecular entity, first indication; retail pharmacy channel; therapeutic area derived from the indication. The forecast states which payer population it represents, with the caveat that the historical analog data in our project are Medicare/Medicaid-based and may differ from commercial behavior.

Two questions for leadership

- **How big can the launch be (ceiling)? Epi funnel**
    - Prevalent population → diagnosed → treated → eligible → peak share → M
    - Line of therapy: first-line M is much larger than third-line M
    - Peak share: default from order of entry (Zipf), checked against analog peaks
- **How fast can we get to the ceiling (trajectory)? Analogs**
    - Borrow uptake shape (Bass p and q) from previous launches
    - Novelty: first in class, or entering a competitive market
    - Line of therapy and competition shape which launches are comparable

Data limits to state up front

- Distribution channel (pharmacy-based vs. medical-based): our data covers pharmacy-dispensed drugs only.
- For novel competitive drugs, distribution data may not be available, because manufacturers may keep it private.

Forecaster Journey

1. Define the drug

The forecaster enters the brand profile:

- Indication and target patient definition (e.g., adults with moderate-to-severe plaque psoriasis who are candidates for systemic therapy and have not yet received a biologic)
- Class (narrow and broad, e.g., TYK2 inhibitor / JAK family)
- Route of administration/formulation (oral, self-injected) - check terms
- Line of therapy
- Order of entry (rank among entrants) and expected eventual number of entrants
    - Define what "first" means: first in class, first for the indication, or first at this line
    - Classes can compete with each other and can expand
- Competitors at launch
    - Number of competitors
    - How does the drug compare on efficacy and safety to the standard of care
- Dosing frequency and days supply - one of the most important
    - Important: how it compares to current alternatives
- Launch date

Output: a saved brand profile that drives the analog search and the epi funnel.

2. Find comparable launches (agent-assisted)

Analog data (built ahead of time, not at query time):

- Build the analog pool from the data schema for analogs (one analog = brand × route group, launches from roughly 2010 onward, since SDUD managed-care records start in 2010)
- Structured sources (NDC directory, Drugs@FDA, SDUD) fill the tabular fields
- The agent reads approved sources (original approval labels, ClinicalTrials.gov, press releases/SEC filings for launch dates) to fill fields that aren't tabular: indication, dosing, class, line of therapy, competitors at launch, launch date
    - Each field is saved with a quote, a link, and a human-approved flag
    - A person reviews the extracted fields once

Analog agent (at query time):

- Code applies **hard filters**: novel NDA/BLA, matching route, launch window, enough SDUD data
- Code scores similarity on the structured features (class, order of entry, route, dosing)
- The agent re-ranks the shortlist, using line of therapy, competition, and other launch characteristics
- An agent proposes about 10 analog drugs, ranked, each with written reasoning. The forecaster reviews and approves the final set of 3-5 analog drugs.

Output: an approved set of analogs, **each with a summary card** (match table, rationale with evidence, distortion flags, data quality).

3. Understand the potential market

Ideally, the user would perform an epidemiology funnel study to define the distribution of each stage: Prevalent population → diagnosed → drug-treated → eligible → peak brand share → M (peak patients on therapy)

Simplified version: Develop market size using Zipf's Law + some adjustments - order of entry vs. share of the market. Adjust the values by deciding how steep the curve falls (alpha) and the number of eventual entrants.

- Inputs: order of entry (from step 1), number of eventual entrants (a range), alpha (a range)
- Optional differentiation adjustment, with a written reason
- Revisit peak share after the analogs are approved

Output: a range for the maximum potential patient population.

4. Review historical uptake

For each approved analog, the user sees its launch trajectory: TRx over time, then normalized uptake, then a fitted Bass curve.

Uptake curve: the share of a drug's eventual peak TRx reached in each quarter after launch

- Normalized curves are overlaid, with the spread shown as a band.
- A small table lists each analog's fitted (p, q) and fit quality, flagging weak fits.
- This is a review step. If an analog looks like an outlier or fits poorly, the user can drop it and the (p, q) spread updates.

Output: the range of uptake speeds and shapes implied by history.

The agent writes a summary of the analogs:

- Similarities and differences between analogs and the new drug
- Which analogs to weight more, and which components to take
- A recommended weighting for the uptake curve, with the reasoning (the weights and curves themselves are computed by code)

5. Illustrative sample curve (demo only)

- Combine the similarity-weighted uptake curve band with a placeholder M, using simple hard-coded assumptions
- No access scenarios, fills conversion, or Monte Carlo

- Labeled on screen as “illustrative,” with the limitations listed next to it: placeholder M, Medicaid-derived shapes, few analogs, no access or persistence modeling, no validation of the forecast itself

Output: a sample curve with a band to show how the analog work feeds a forecast.